Finite Element Analysis of Anisotropic Optical Waveguide with Arbitrary Index Profile

Marcos A. R. Franco, Angelo Passaro
Centro Técnico Aeroespacial - CTA/IEAv, Rod. Tamoios, km.5,5 - CEP. 12228-840, São José dos Campos, SP, Brazil,

José R.Cardoso
LMAG - PEA - EPUSP - Av. Prof. Luciano Gualberto, Trav.3, no.158 - CEP05508-900 - São Paulo, SP, Brazil

José M. Machado
Depto. de Ciências da Computação e Estatística IBILCE - UNESP - CP 136 - CEP.: 15054-000 - S. J. Rio Preto - SP - Brazil

Abstract – **This work presents the application of a scalar finite element formulation for E^x (TE-like) modes in anisotropic planar and channel waveguides with diagonal permittivity tensor, diffused in both transversal directions. This extended formulation considers explicitly both the variations of the refractive index and their spatial derivates inside of each finite element. Dispersion curves for E^x modes in planar and channel waveguides are shown, and the results compared with solutions obtained by other formulations.**

Index terms – **Finite element methods, nonhomogeneoulsy loaded waveguides, optical strip waveguides, anisotropic media.**

I. Introduction

Diffused optical waveguides are used in many integrated optical circuits, and design tools for these waveguides present a great technological importance. The computer simulation of diffused optical waveguides has been extensively applied in the last two decades for the study of several optical devices. The simulation works frequently found in the literature are based on different mathematical methods: FEM, FD, Effective Index Method, Variational Method, etc. In order to improve the quality of the design tools, a good comparison on the accuracy of these mathematical methods is necessary. Notwithstanding the several works published about the simulation of isotropic diffused waveguides [1]-[8], results concerning the study of anisotropic diffused waveguide are not very often found [9]-[15].

We presented in a previous work the scalar finite element (FE) formulation for anisotropic diffused waveguides in which the refractive index is nonhomogeneous in only one direction [9]. That formulation is suitable for APE (annealed-proton exchanged) $LiNbO_3$ waveguides. In this paper, we extend the scalar FE formulation previously mentioned for three refractive indexes which vary continuously along the transversal directions. This formulation can be applied to study waveguides in which Ti is diffused in a $LiNbO_3$ substrate.

In order to validate the formulation, six cases were studied: a homogeneous channel waveguide, an isotropic diffused planar waveguide, two isotropic diffused channel waveguides, and two anisotropic diffused channels, a APE $LiNbO_3$ and a Ti:$LiNbO_3$ waveguide. The corresponding results were compared with solutions obtained by vector formulations, analytical methods and other scalar formulations.

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II. Finite Element Formulation

For anisotropic dielectrics in which the relative permittivity tensor is diagonal

$$\bar{\varepsilon}_R = \begin{bmatrix} n_x^2(x,y) & 0 & 0 \\ 0 & n_y^2(x,y) & 0 \\ 0 & 0 & n_z^2(x,y) \end{bmatrix},$$

where n_x, n_y, n_z are the refractive indexes, the scalar wave equation for E^x modes ($E_y = 0$ for TE approximation) is written as follows [9]:

$$\frac{\partial^2}{\partial x^2}(n_x^2\ E_x) + n_z^2 \frac{\partial^2 E_x}{\partial y^2} + n_z^2 \frac{\partial}{\partial x}\left(\frac{1}{n_z^2}\right)\frac{\partial}{\partial x}(n_x^2\ E_x) - \beta^2 n_z^2 E_x + k_0^2\ n_x^2\ n_z^2\ E_x = 0\,. \quad (1)$$

Applying the Weighted Residual Method to (1) yields:

$$[F]\{E_x\}^T = n_{eff}^2\,[M]\{E_x\}^T \quad (2)$$

where $n_{eff} = \beta\ /\ k_0$ is the effective index, and k_0 is the free-space wavenumber. The matrices [F] and [M] for triangular finite elements are :

$$[M] = 2\Delta k_0^2 \iint_{\zeta_1\,\zeta_2} n_z^2 \{N\}^T\{N\} d\zeta_1\, d\zeta_2\,, \quad (3a)$$

$$[F] = [F_1] - [F_2] - [F_3] + [F_4]\,, \quad (3b)$$

where:

$$[F_1] = 2\Delta k_0^2 \iint_{\zeta_1\,\zeta_2} n_x^2\, n_z^2 \{N\}^T\{N\} d\zeta_1\, d\zeta_2$$

$$[F_2] = \sum_{i,j=1}^{3}\left(\frac{b_i\, b_j}{2\Delta}\iint_{\zeta_1\,\zeta_2}\left(n_x^2 \frac{\partial\{N\}^T}{\partial\zeta_i}\frac{\partial\{N\}}{\partial\zeta_j} + \delta_x \frac{\partial\{N\}^T}{\partial\zeta_i}\frac{\partial n_x^2}{\partial\zeta_j}\{N\}\right) d\zeta_1\, d\zeta_2\right)$$

$$[F_3] = \sum_{i,j=1}^{3}\left(\frac{c_i\, c_j}{2\Delta}\iint_{\zeta_1\,\zeta_2}\left(n_z^2 \frac{\partial\{N\}^T}{\partial\zeta_i}\frac{\partial\{N\}}{\partial\zeta_j} + \delta_z \{N\}^T \frac{\partial n_z^2}{\partial\zeta_i}\frac{\partial\{N\}}{\partial\zeta_j}\right) d\zeta_1 d\zeta_2\right),$$

$$[F_4] = \sum_{i,j=1}^{3}\left(\frac{b_i\, b_j}{2\Delta}\iint_{\zeta_1\,\zeta_2}\left(\delta_z\, n_z^2 \frac{\partial g_z^2}{\partial\zeta_i}\right)\left(\delta_x \frac{\partial n_x^2}{\partial\zeta_j}\{N\}^T\{N\}\right) d\zeta_1\, d\zeta_2\right) + \sum_{i,j=1}^{3}\left(\frac{b_i\, b_j}{2\Delta}\iint_{\zeta_1\,\zeta_2}\left(\delta_z\, n_z^2 \frac{\partial g_z^2}{\partial\zeta_i}\right)\left(n_x^2\{N\}^T \frac{\partial\{N\}}{\partial\zeta_j}\right) d\zeta_1\, d\zeta_2\right)$$

Δ is the triangle area, $\{N\}$ is a row matrix representing a complete set of base functions, $\{\ \}^T$ stands for a transposed matrix, ζ_i are the independent homogeneous coordinates and b_i and c_j are the homogeneous system coordinate-transformation coefficients. Apart from the terms containing the refractive index partial derivative, this is the same approach used by Koshiba for 1D anisotropic planar guides [14] and for isotropic channel waveguides [6]. In $[F_2]$, $[F_3]$ and $[F_4]$ the parameters δ_x and δ_z assume either the value 1 for diffused index in the x and z direction, respectively, or zero for constant index. In anisotropic homogeneous regions, the matrices (3) are reduced to the matrices presented in [16], and for isotropic homogeneous regions to the matrices presented in [17]. The matrices $[F_2]$, $[F_3]$ and $[F_4]$ are sparse and nonsymmetric due to the presence of terms with $dn^2/d\zeta$.

The state variable E_x and the refractive indexes are modeled using the nodal approximation:

$$E_x = \{N\}\{E_x\}^T, \tag{4}$$

$$n_k^2 = \{N\}\{n_k^2\}^T, k = x, z\ , \tag{5}$$

and

$$g_z^2 = \frac{1}{n_z^2} = \{N\}\{g_z^2\}^T \tag{6}$$

All integrals in (3) were previously calculated by analytical integration, using the same approach described in [18].

III. Homogeneous Isotropic Channel Waveguide

The homogeneous rectangular waveguide was simulated and the results for fundamental E^x mode compared with other numerical methods (Fig.1). This simple example was used to test the formulation for homogeneous waveguides. The Fig.1 shows the good agreement with the vector integral equation method (VIE) and the effective index method (EIM) [7]. The Marcatili method [19] obtains lower propagation constant near the cutoff.

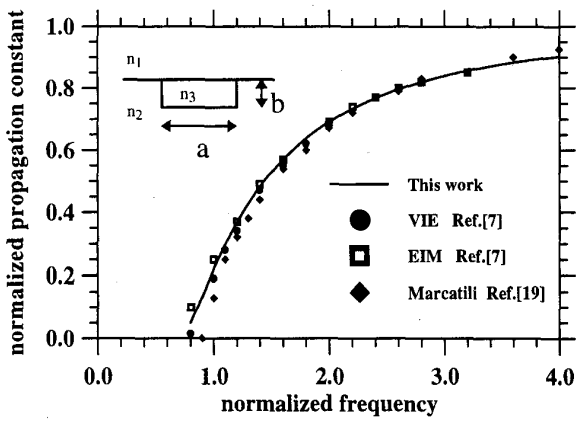

Fig. 1 Dispersion curves for isotropic homogeneous channel waveguide. The normalized frequency is $(k_o\, b / \pi)\ (n_3^2 - n_2^2)^{1/2}$ and the normalized propagation constant is $(n_{eff}^2 - n_2^2) / (n_3^2 - n_2^2)$. The refractive index are n_1=1.0, n_2=1.43 and n_3=1.50.

IV. Planar Diffused Isotropic Waveguide

A planar optical waveguide with exponential index profile was used to verify the validity of the present formulation. Fig.2 shows the effective index for the lowest three E^x modes. The results agree with both the exact and numerical solutions, given in [6] and represented by the points.

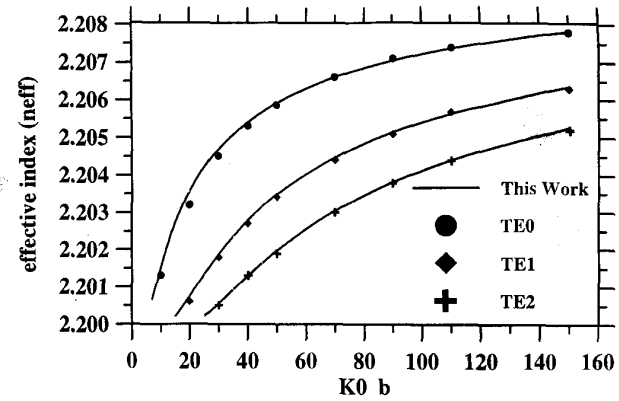

Fig. 2 Dispersion curves for planar isotropic diffused optical waveguide, where b is the diffusion depth and $n(y) = 2.20 + 0.01 \exp(-|y| / b)$.

V. Isotropic Diffused Channel Waveguide

A rectangular cross section waveguide was considered in which a circular diffusion was performed only within a rectangular core. (Fig. 3).

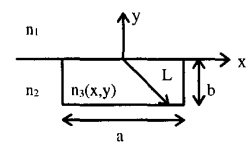

Fig. 3 channel waveguide geometry.

The refractive index for the circular diffusion is given by [8]

$$n_3 = n_2 + \frac{n_2 - n_{3m}}{L^2}\left(x^2 + y^2 - L^2\right), \tag{7}$$

in which n_2 is the value for the index in the substrate, and n_3, n_{3m} are the refractive index and the maximum refractive index inside the waveguide respectively. L is the length of the straight line from the origin to a point on the core boundary which intersects a point P(x,y). If the rectangular coordinates of P satisfy $|y| \geq |x|$ then:

$$L = \sqrt{b^2 + x^2}\ , \tag{8}$$

otherwise

$$L = \sqrt{(a/2)^2 + y^2}\ , \tag{9}$$

where **a** and **b** are the width and the height of the rectangular cross section, respectively. The entire domain (including the air region) was discretized into a mesh of 5680 first order triangles, corresponding to 2800 nodes. In order to reach a better description of the modes near the cutoff condition, this mesh was well refined inside the core and in all substrate regions where the field strength is not negligible. The values taken for the simulation were $n_2 = 1.44$, $n_{3m} = 1.5$ and $n_1 = 1.0$. Far from the cutoff region, the results obtained by our implementation for an isotropic diffused waveguide are in good agreement with the results presented in the literature (Figs. 4 and 5). Near the cutoff region, discrepancies appear between all simulations. The curves shown in Fig. 4 allow comparison of the results obtained by a vector integral equation method [7], a vector finite element formulation (VFEM) [8] and a scalar finite element formulation (SFEM) [6].

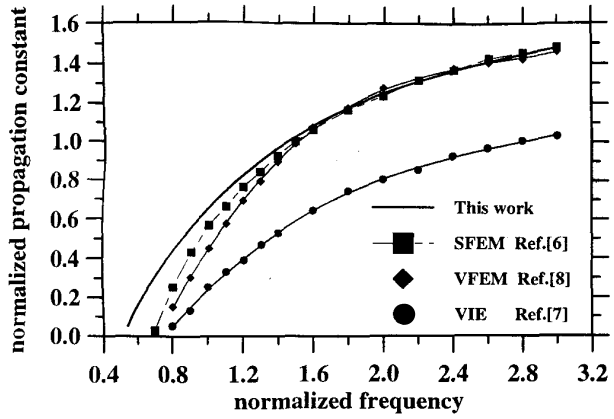

Fig. 4 Dispersion curves for the E^x_{11} mode in a waveguide with circular index profile. The normalized frequency is $(k_o\, b / \pi)\ (n_{3av}^2 - n_2^2)^{1/2}$ and the normalized propagation constant is $(n_{eff}^2 - n_2^2) / (n_{3av}^2 - n_2^2)$, where $n_{3av} = 1.47$ is the averaged index in core region.

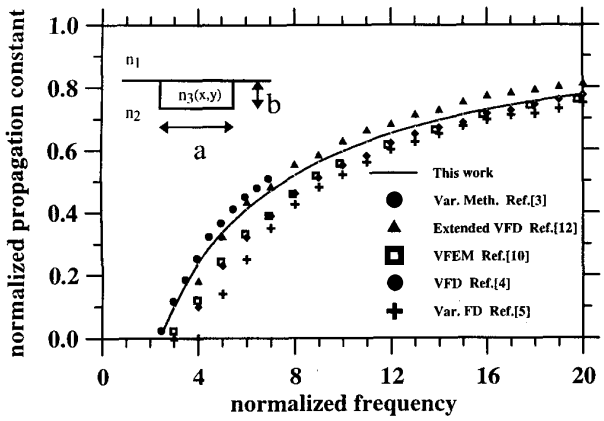

Fig. 5 Dispersion curves for E^x_{11} mode in channel isotropic diffused waveguide with Gaussian-Gaussian index profile. $n_3 = n_2\,(1 + 0.05\, f(x,y))$, $a/b = 1$. The normalized frequency is $(k_o\, b / \pi)\ (n_{3m}^2 - n_2^2)^{1/2}$ and the normalized propagation constant is $(n_{eff}^2 - n_2^2) / (n_{3m}^2 - n_2^2)$ with $n_1 = 1.0$, $n_2 = (2.1)^{1/2}$ and $n_{3m} = 1.05\, n_2$.

Moreover, it should be noticed in Fig. 5 that the scalar formulation used in this work reproduces very well the results obtained from other methods, namely: the Variational Method (VM) [3], the Vector FE [10] and Vector FD (VFD) [4] Methods, the Extended Vector FD[12] and the Variational FD (Var.FD) [5] Methods. For a region close to the cutoff our results, shown in Fig. 5, are in good accordance with the results presented in [3] which also uses a refined mesh in the region of interest.

VI. Anisotropic Diffused Channel Waveguide

A. Annealed proton-exchanged waveguide

Annealed proton-exchanged (APE) $LiNbO_3$ optical waveguides have the property of single polarization due to APE process changes only the extraordinary refractive index.

The index change after proton exchange (PE) process is approximated by a initial step index ($\Delta n_{PE} = 0.12$), subsequently the annealing process was simulated by a 2D linear anisotropic diffusion equation solved by the FEM.

The extraordinary index change was related to the proton concentration in the substrate as follows [20]:

$$n_e(C) = n_{es} + \Delta n_e\left[1 - \exp(-11\, C)\right], \tag{8}$$

where C is the normalized concentration of protons (0<C<1), n_{es} is the extraordinary refractive index of the $LiNbO_3$ and Δn_e is the maximum refractive index change induced by the proton exchange (PE) process.

Fig. 6 shows the dispersion curves for the lowest four E^x modes in x-cut $LiNbO_3$ guide. The PE and anneal times are fifteen minutes at 190° C and four hours at 360° C, respectively.

The dispersion of the refractive index was not considered in this example. A detailed analysis for this case was presented in a previous work [9].

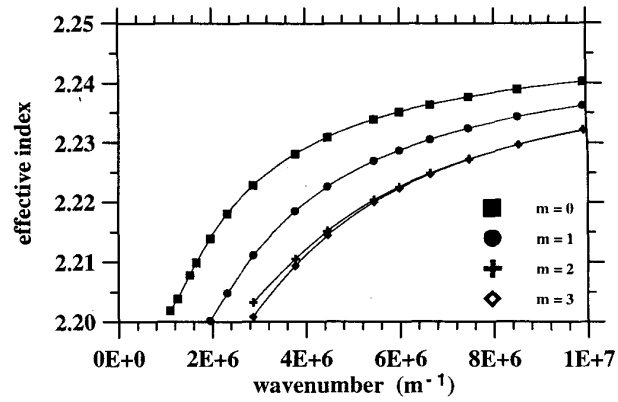

Fig. 6: Dispersion curves for lowest four E^x_m modes in APE x-cut $LiNbO_3$ waveguide. For 360°C the anneal diffusion constants are Da(x cut) = 0.92 $\mu m^2/h$, Da(z cut) = 0.77 $\mu m^2/h$, and $\lambda o = 0.6328\ \mu m$.

B. Ti diffused $LiNbO_3$ waveguide

For Ti:$LiNbO_3$ channel waveguides, the refractive index in the diffused region follows [21]:

$$n_{e,o}^2(x,y,\lambda) = n_{b_{e,o}}^2 + \left[\left(n_{b_{e,o}} + \Delta n_{s_{e,o}}\right)^2 - n_{b_{e,o}}^2\right] \exp\left(-\frac{y^2}{d_y^2}\right) f\left(\frac{2x}{W}\right), \tag{10}$$

where

$$f\left(\frac{2x}{W}\right) = \frac{1}{2}\left\{\mathrm{erf}\left[\frac{W}{2d_x}\left(1 + \frac{2x}{W}\right)\right] + \mathrm{erf}\left[\frac{W}{2d_x}\left(1 - \frac{2x}{W}\right)\right]\right\}, \tag{11}$$

e, **o** denote the extraordinary and ordinary rays respectively, x and y are the coordinates for a point in the substrate, W is the initial width of the Ti strip, d_x and d_y are the diffusion width and depth respectively, n_b is the substrate refraction index and Δn_s stands for the variation of the surface index with the wavelength.

In addition, $\Delta n_{sc,o}$ is given in terms of H (the initial thickness of the Ti strip) and some fitting parameters [22]:

$$\Delta n_{s_{e,o}}(\lambda) = \left[B_0(\lambda) + B_1(\lambda)\frac{H}{d_{y_{e,o}}}\right]\left(\frac{H}{d_{y_{e,o}}}\right)^{\alpha_{e,o}}, \tag{12}$$

$\alpha_e = 0.83,\ \alpha_o = 0.53,$

$B_{0e}(\lambda) = 0.385 - 0.430\lambda + 0.171\lambda^2,$

$$B_{1e}(\lambda) = 9.130 + 3.850\lambda - 2.490\lambda^2,$$
$$B_{0o}(\lambda) = 0.0653 - 0.0315\lambda + 0.0071\lambda^2,$$
$$B_{1o}(\lambda) = 0.4780 + 0.4640\lambda - 0.3480\lambda^2.$$

The simulations were performed for a guide built in a x-cut crystal with initial Ti stripe thickness H=100 nm, λ=1.523 μm, T=1050 °C and t=8.5 h of diffusion time. Using the diffusion parameters obtained from [21] and [22] the follow input data were derived: d_x=4.60 μm, d_y=4.00 μm, d_{yo}=6.23 μm, d_{ye}=4.98 μm, n_{bo}=2.2125, n_{be}=2.1383, Δn_{so}=0.00446 and Δn_{se}=0.01217.

Fig.7 shows the calculated effective index for E^x_{11} mode and the mode sizes W_x and W_y (full width at half maximum field intensity) versus the initial width of the Ti stripe. The behaviour of these curves are in good agreement with the results presented in [21]. These results were not included in the figure since their curves were obtained for a c-cut crystal.

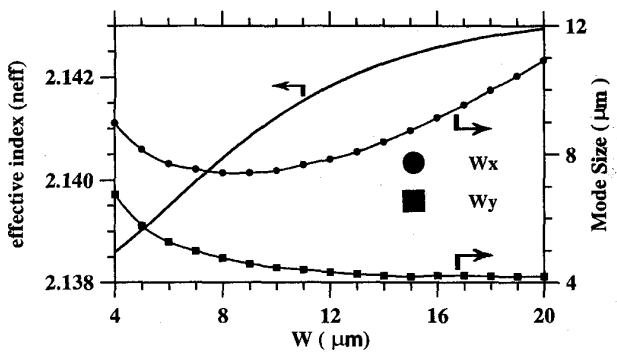

Fig. 7 Effective index (n_{eff}) and mode sizes W_x and W_y versus the initial width of the Ti stripe.

VIII. Conclusions

We presented a comparison of the results obtained from various formulations proposed in the literature for the propagation modes calculation in optical waveguides of technological importance.

The characterization of waveguides near the cutoff region is important for the design of monomode devices, but in this region discrepancies appear between all simulations. Unfortunately, there are few comparisons in the literature between experimental and simulated results.

The comparisons showed that the scalar formulation seems to be as reliable as other methods commonly used to study the propagation modes in diffused optical waveguides. This information is important to develop reliable software tools for the analysis of complex integrated optical devices. The scalar formulations consume less memory and CPU time than the vector formulations allowing the analysis of these complex devices even in low cost computers.

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